

SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 3 – CODING CHALLENGE QUESTIONS

Course Code:	CSA1220	Course Title:	Computer Architecture
CO Assessed:	CO1 – Precision (BL4)	Assessment:	Coding challenge questions
Weightage:	40%	Total Marks:	100
CO1 Statement:	Analyse the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking		
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Section / Batch:	CSE(AI) / 2ND YEAR	Date:	03/08/2026

Instructions to Students

- Analyze the given scenario and identify the problem requirements before developing the solution.
- Design the algorithm or flowchart and select the appropriate 8085 instructions, registers, and addressing modes required for implementation.
- Develop and execute the 8085 Assembly Language Program (ALP) using a simulator or trainer kit to solve the given problem.
- Verify and validate the output by testing the program with the given input data and ensuring the expected results are obtained.
- Interpret the program execution by explaining the logic used, the role of registers, memory locations, flags, and the final output generated.

QUESTIONS

1. A smart factory stores the hourly production count of eight machines in consecutive memory locations. The supervisor wants to identify the machine with the highest production and determine whether the total production exceeds the daily target stored in another memory location, Design and implement an 8085 ALP to:
 - Calculate the total production.
 - Find the highest production value.
 - Compare the total with the target value.
 - Store a status flag indicating whether the target has been achieved.
2. An automated chemical plant records ten temperature readings. A reading is considered abnormal if it is greater than the predefined threshold stored in memory. The controller must count the number of abnormal readings and identify the highest recorded temperature. Write an 8085 ALP that
 - Scans all sensor readings.
 - Counts the number of abnormal readings.
 - Finds the maximum temperature.
 - Stores both results in memory.
3. An educational institution stores marks of ten students in memory. A student is considered successful if the mark is greater than or equal to the pass mark stored in memory. Develop an 8085 ALP to
 - Count the number of students who passed.
 - Find the highest mark.
 - Calculate the total marks obtained by all students.
 - Store all results in separate memory locations.

Code of Conduct

I LOKESH S (192572151) (Name / Reg No) certify that this submission is my original work and that I

have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: LOKESH S

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly understands problem constraints, edge cases, and competitive requirements	Understands problem with minor gaps in constraints	Partial understanding; misses key constraints	Misinterprets problem or constraints
Algorithm	25%	Selects optimal algorithm with best time and space complexity for competitive constraints	Uses efficient algorithm with minor scope for improvement	Uses basic/inefficient approach	No clear algorithmic approach
Code Implementation	20%	Writes correct, efficient, and clean code that passes all constraints	Code works with minor errors or inefficiencies	Partially correct code; fails for some cases	Code does not execute or gives incorrect results
Competitive Performance	20%	Solves problem within time limits and handles large inputs effectively	Solves within acceptable time with minor delays	Struggles with time limits or larger inputs	Unable to solve within time constraints
Test Case Handling	15%	Handles all test cases including edge and hidden cases	Handles most test cases	Misses important edge cases	Fails on multiple test cases

Marks Evaluation:

Q. No	Problem Understanding (20)	Algorithm (25)	Code Implementation (20)	Competitive Performance (20)	Test case handling (15)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	

2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 20	/ 25	/ 20	/ 20	/ 15	/ 100

Course Coordinator Faculty Incharge

1. PRODUCTION COUNT OF EIGHT MACHINES

LXI H, 8050H ; HL points to first production count
MVI C, 08H ; Number of machines

MVI D, 00H ; D:E stores 16-bit total
MVI E, 00H

MVI B, 00H ; B stores maximum production
MVI A, 01H

STA 8063H ; Initially machine number = 1
MVI A, 01H
STA 8065H ; Temporary current machine number

LOOP: MOV A, E ; Add current production to total
ADD M
MOV E, A

MVI A, 00H
ADC D
MOV D, A

MOV A, M ; Compare current production with maximum
CMP B
JC NEXT
JZ NEXT

MOV B, A ; Update maximum production
LDA 8065H
STA 8063H ; Store machine number

NEXT: INX H ; Point to next machine
LDA 8065H
INR A
STA 8065H

DCR C
JNZ LOOP

MOV A, E ; Store total low byte
STA 8060H

MOV A, D ; Store total high byte
STA 8061H

MOV A, B ; Store maximum production
STA 8062H

; Compare total with 16-bit target

LDA 8059H ; Load target high byte
MOV C, A
MOV A, D ; Total high byte
CMP C
JC NOTACH ; Total high < target high
JNZ ACHIEVE ; Total high > target high

LDA 8058H ; High bytes equal, compare low bytes
MOV C, A
MOV A, E
CMP C
JC NOTACH

ACHIEVE:

MVI A, 01H
STA 8064H
HLT

NOTACH:

MVI A, 00H
STA 8064H
HLT

RESULT

The program calculates the 16-bit total production, finds the highest production and corresponding machine number, compares the total with the target, and stores the status flag.

2. TEMPERATURE MONITORING IN A CHEMICAL PLANT

LXI H, 8050H ; HL points to first temperature
MVI C, 0AH ; Ten temperature readings

LDA 805AH ; Load threshold value
MOV D, A ; D stores threshold

MVI B, 00H ; B stores abnormal count
MVI E, 00H ; E stores maximum temperature

LOOP: MOV A, M ; Load current temperature

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CMP E      ; Compare with current maximum
JC CHECK
JZ CHECK
MOV E, A   ; Update maximum
```

CHECK:

```
MOV A, M
CMP D      ; Compare reading with threshold
JC NEXT    ; Reading < threshold
JZ NEXT    ; Reading = threshold
```

```
INR B      ; Reading > threshold
            ; Increment abnormal count
```

```
NEXT: INX H      ; Point to next temperature
DCR C
JNZ LOOP
```

```
MOV A, B      ; Store abnormal count
STA 8060H
```

```
MOV A, E      ; Store maximum temperature
STA 8061H
```

```
HLT
```

RESULT

The program scans all ten temperature readings, counts the readings greater than the threshold, finds the maximum temperature, and stores both results in memory.

3. PROCESSING MARKS OF TEN STUDENTS

LXI H, 8050H ; HL points to first student's mark
MVI C, 0AH ; Number of students

LDA 805AH ; Load pass mark
MOV D, A ; D stores pass mark

MVI B, 00H ; B stores number of passed students
MVI E, 00H ; E stores highest mark

MVI A, 00H
STA 8062H ; Initialize total low byte
STA 8063H ; Initialize total high byte

LOOP: MOV A, M ; Load current student's mark

; Find highest mark

CMP E

JC CHECKPASS

JZ CHECKPASS

MOV E, A ; Update highest mark

; Check whether student passed

CHECKPASS:

MOV A, M

CMP D

JC ADDTOTAL ; Mark is less than pass mark

INR B ; Mark is equal to or greater than pass mark

; Add mark to 16-bit total

ADDTOTAL:

LDA 8062H ; Load total low byte

ADD M

STA 8062H

LDA 8063H ; Load total high byte

ACI 00H ; Add carry to high byte

STA 8063H

INX H ; Point to next mark

DCR C

JNZ LOOP

MOV A, B ; Store number of passed students

STA 8060H

MOV A, E ; Store highest mark

STA 8061H

HLT

RESULT

The program counts the students whose marks are greater than or equal to the pass mark, finds the highest mark, calculates the 16-bit total of all ten marks, and stores the results in separate memory locations.